

9	C	Loss in G.P.E. by Y = gain in G.P.E. by X + gain in K.E. by X and Y $3.0 \times 9.81 \times 3.0 \sin 40^\circ = 2.0 \times 9.81 \times 3.0 \sin 30^\circ + \frac{1}{2}(2.0 + 3.0)v^2$ $v \approx 3.3 \text{ m s}^{-1}$
10	D	For mass m: